

PH.D. EXAM in ANALYSIS

may 11, 1999.

(I) State the following theorems:

- 1.) Hahn - Banach
- 2.) The Egorov theorem
- 3.) The Vitali convergence theorem
- 4.) The Radon - Nikodym theorem
- 5.) The Fubini theorem
- 6.) The monotone convergence theorem
- 7.) The completeness theorem for $L_F^1(\mu)$.
- 8.) The equality $\|gm\| = \|g\| \|m\|$.

(II) Prove one of the theorems 3 or 7.

(III) Solve the following problems, where (X, Σ, μ) is a measure space and F is a Banach space.

- 1.) If $f_n \rightarrow f$ in $L_F^1(\mu)$ and $f_n \rightarrow g, \mu$ -a.e., then $f = g, \mu$ -a.e.
- 2.) Let $\mathcal{F} \subset \Sigma$ be a σ -algebra. Prove the existence of the conditional expectation $E(f|\mathcal{F})$ for $f \in L_F^1(\mu)$.
- 3.) Let $\mathcal{R}$ be a ring generating Σ and $f \in L_F^1(\mu)$. Prove that if $\int_A f d\mu = 0$ for every $A \in \mathcal{R}$, then $f = 0, \mu$ -a.e.
- 4.) Let $(X, \mathcal{F})$ and $(Y, \mathcal{T})$ be measurable spaces, $\mu: \mathcal{F} \rightarrow \mathbb{R}$ and $\nu: \mathcal{T} \rightarrow \mathbb{R}$, σ -additive measures. Prove the existence on the product measure $\mu \times \nu: \mathcal{F} \times \mathcal{T} \rightarrow \mathbb{R}$ satisfying $\mu \times \nu(A \times B) = \mu(A)\nu(B)$ for $A \times B \in \mathcal{F} \times \mathcal{T}$ and $|\mu \times \nu| = |\mu| \times |\nu|$.
- 5.) If $f \in L_F^1(\mu)$ there is a sequence (f_n) of $\mathbb{R}$ -step functions

Note.. Write complete computations and explanation
Do not use arrows or other unnecessary signs. Use
words for explanations.
Write complete sentences.